Instruction Tuning and Direct Preference Optimization on Qwen2.5-0.5B

Programming Assignment 2 — Final Report

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Repository: https://github.com/fazalreh/small-model-alignment

Abstract

We post-train `Qwen/Qwen2.5-0.5B` with a from-scratch supervised fine-tuning (SFT) loop on 30k UltraChat examples, then run two from-scratch DPO experiments from the same SFT checkpoint: one on 15k single-turn HH-RLHF pairs and one on 15k UltraFeedback pairs. A third track tunes `SmolLM2-360M-Instruct` with TRL's `DPOTrainer` and LoRA toward a Gordon-Ramsay-style persona. All models are compared on fixed prompt sets with blinded, order-swapped pairwise judgments from `google/gemini-3.6-flash` through Kaggle's model proxy. SFT lowers held-out response NLL from 1.396 to 1.331 and makes the model stop properly (65% of base-model answers hit the 512-token cap). UF-DPO reaches 0.775 validation reward accuracy. For HH, the guide learning rate (10^{-6}) barely changes refusal behavior, so we sweep $\{1, 3, 10\} \times 10^{-6}$ and select 3×10^{-6} by validation reward accuracy (0.654). It raises refusal and redirection on harmful prompts, while 10^{-5} over-optimizes into incoherent over-refusal. Judged against SFT, UF-DPO scores 52.2% (95% CI 48%–57%) on AlpacaEval and HH-DPO scores 69.0% (95% CI 64%–73%) on the helpful/harmless set (ties counted as half). HH-DPO's gain comes from refusing actionable harm without over-refusing benign requests. UF-DPO mainly lengthens answers, which the length-neutral judge does not reward. LoRA-DPO shows a clean style–quality trade-off: persona win rate 55/85/94% against quality 41/24/14% across the sweep.

1 Repository, models, and data

Code. The training math (batching, masking, response log-probabilities, SFT and DPO losses) lives in `sma/core.py`, and `submission_adapter.py` wraps it for the private grader. The loops are `sma/train_sft.py` and `sma/train_dpo.py`; neither uses HF `Trainer` or TRL. The 7 public tests and 15 additional edge-case tests pass (truncation that removes EOS, `pad_token_id=eos_token_id`, missing pad/EOS tokens, row order, brute-force log-prob checks, gradient-accumulation equivalence). We build on DPO [1], LoRA [9], Qwen2.5 [10], SmolLM2 [11], UltraChat [4], HH-RLHF [3], UltraFeedback [5], AlpacaEval [6, 7], and the MT-Bench pairwise-judging protocol [8]; the overall pipeline follows [2]. Software (logged in each `run_config.json`): torch 2.6.0 (CUDA 12.4), transformers 4.51.3, datasets 3.6.0, TRL 0.19.1, PEFT 0.15.2.

Prompt format. The same template is used for SFT, both DPO runs, and evaluation: Qwen ChatML with the system message "You are a helpful assistant." (`sma/prompts.py`). The response is followed by the tokenizer EOS `<|endoftext|>`, and generation stops on `<|im_end|>` or `<|endoftext|>`. Prompt and response are tokenized separately and concatenated, EOS is appended *before* truncation to 512 tokens, rows are right-padded, and all masks are derived from true row lengths (important because Qwen's pad id equals its EOS id).

Seeds and decontamination. Seed 42 is used for every shuffle, the evaluation subset, data order, and A/B order. Training prompts that exactly match (after whitespace and case normalization) an evaluation instruction are removed.

Table 1: Data actually used. Counts are from `outputs/data/data_stats.json`.

Track	Source	Filtering	Train / val
SFT	UltraChat-200k (`HuggingFaceH4/ultrachat_200k`)	first user/assistant turn; prompt ≤256 tok; prompt+response+EOS ≤512 tok (30,000 kept of 59,174 scanned)	30,000 / 1,000
HH-DPO	`Anthropic/hh-rlhf`, harmless-base + helpful-base	single-turn only (24,971 of 86,372); chosen ≠ rejected; prompt ≤256 tok; 50/50 harmless/helpful	15,000 / 1,000
UF-DPO	UltraFeedback (`HuggingFaceH4/ultrafeedback_binarized`)	single-turn; score margin ≥1 (drops 18,447 near-ties); prompt ≤256 tok (33,474 eligible)	15,000 / 1,000
LoRA-DPO	provided `ramsay_style_preferences.json`	seeded 90/10 split	198 / 22

Evaluation sets: AlpacaEval subset of 300 of 805 prompts drawn with `random.Random(42).sample` (`eval_sets/alpaca_eval_300_seed42.json`); the provided HH set (300 prompts, 10 categories × 30); our persona set of 40 prompts (10 each: cooking, coding, general help, safety-sensitive).

2 Experimental configuration

For each pair (x, y_w, y_l) we minimize the DPO objective, with response-only summed log-probabilities:

$$\mathcal{L}_{\text{DPO}} = -\log \sigma\Big(\beta\big[(\log \pi_\theta(y_w \mid x) - \log \pi_\theta(y_l \mid x)) - (\log \pi_{\text{ref}}(y_w \mid x) - \log \pi_{\text{ref}}(y_l \mid x))\big]\Big), \tag{1}$$
$$r_w = \beta\big(\log \pi_\theta(y_w \mid x) - \log \pi_{\text{ref}}(y_w \mid x)\big), \qquad r_l = \beta\big(\log \pi_\theta(y_l \mid x) - \log \pi_{\text{ref}}(y_l \mid x)\big).$$

Reward accuracy is $\Pr[r_w > r_l]$ (strict) and the margin is $\mathbb{E}[r_w - r_l]$. The SFT loss is the *token mean* of response NLL. With gradient accumulation it is normalized by the response-token count of the whole accumulation window, so the accumulated gradient equals the gradient of the token mean over the full effective batch.

Table 2: Values used for the submitted checkpoints (one H100 80GB on Modal; seed 42 throughout).

	SFT	HH-DPO	UF-DPO	LoRA-DPO
Model	Qwen2.5-0.5B	SFT best	SFT best	SmolLM2-360M-Instruct
Trainable	full (494M)	full	full	LoRA r=16, α=32, drop. 0.05; q,k,v,o,gate,up,down (8.68M, 2.34%)
Optimizer	AdamW (0.9, 0.999)	AdamW	AdamW	AdamW (TRL)
Peak LR, schedule	2×10^{-5}, cosine, 3% warmup	3×10^{-6} constant (sweep $\{1,3,10\}\times10^{-6}$)	10^{-6} constant	sweep $\{5\times10^{-5}, 10^{-4}, 2\times10^{-4}\}$, cosine, 10% warmup
Weight decay / clip	0.1 (not on norms/emb.) / 1.0	0 / 1.0	0 / 1.0	0 / 1.0
Micro × accum = eff.	16 × 2 = 32	4 × 8 = 32 pairs	4 × 8 = 32 pairs	4 × 2 = 8
Precision	bf16 autocast, FP32 weights	same	same	bf16
Max len / prompt	512 / 256	512 / 256	512 / 256	512 / 256
Epochs (steps)	1 (938)	1 (469)	1 (469)	3 (75) / 3 (75) / 5 (125)
β, reference	—	0.1, frozen `sft/best`	0.1, frozen `sft/best`	0.1, adapter disabled
Selection	min val loss (step 900)	max val reward acc (step 400)	max val reward acc (step 400)	final adapter; run chosen by a pre-registered rule (§5.1)
Wall-clock	6.0 min	7.3 min (each run)	12.4 min	∼1 min

Decoding (identical within every comparison). Greedy (`do_sample=False`, so temperature and top-p are unused), `max_new_tokens`=512, `repetition_penalty`=1.1, no length penalty, no n-gram blocking, bf16, left-padded batches. The Qwen models use the ChatML template above; the SmolLM2 models use their own chat template.

Before each full run we inspected 5 sanity-check generations per model (checkpoint path, template, EOS stopping, length). **Judge.** `google/gemini-3.6-flash` at temperature 0, 5 pairs per request, ≥5 s between attempts, ≤3 retries, both A/B orders. The 7 general/safety comparisons were judged in one Kaggle batch run by 6 concurrent workers (one pair file each, each sleeping ≥5 s between its own requests), with Gemini's thinking disabled (`reasoning="none"`) and responses clipped at 4,000 characters. This took 30 min and $2.83 of the free daily quota, with 0 failures and 0 unknown verdicts. The first persona judging used the same settings, but its *style* verdicts were unusable: 39/40 ties, and position consistency as low as 0.43, even when the judge's own reason said one answer "fails the persona". We therefore re-judged only the 3 persona files with `reasoning="low"` and a sharpened style rubric (tone judged independently of quality; "tie" only for equal style). The discarded run is kept in `outputs/judged/persona_reasoning_none/`. Earlier interactive attempts that were cut off by session time-outs left no results and are not reported.

LoRA configuration (Task 5). SmolLM2-360M is a Llama-style decoder, so we adapt *all* linear projections (attention q,k,v,o and MLP gate/up/down). A persona changes word choice and phrasing, which the MLP blocks largely store, so attention-only adapters (the original LoRA default) would limit what DPO can move; we did not ablate this choice. Rank 16 with α=32 (scale 2) adds 8.68M parameters (2.34%). That is plenty for a stylistic shift learned from 198 pairs, while the frozen base keeps the model's knowledge. Dropout 0.05 regularizes against memorizing the tiny dataset. With 198 pairs, a micro-batch of 4 with 2-step accumulation (effective 8) gives 25 updates per epoch. LoRA tolerates a larger learning rate than full fine-tuning, so the epochs and LR were set by the sweep in §5.1. The reference is the base model with the adapter disabled (no second model copy).

3 Parameter choices and deviations

Table 3: Deviations from the guide. R = resource-driven, P = performance-driven, D = design or measurement choice.

Item	Guide	Ours	Type	Reason and observed effect
Hardware	T4×2, fp16+scaler	H100, bf16	R	bf16 has fp32 range, so no loss scaling is needed and no steps are skipped; FP32 master weights unchanged. The fp16 path is kept (`-precision fp16`).
SFT micro-batch	2×16	16×2	R	Same effective batch; window-level token normalization makes the gradient identical (unit-tested).
DPO micro-batch	1×32	4×8	R	Same 32 pairs per step; per-micro-batch means are reweighted.
DPO optimizer	RMSprop/AdamW	AdamW	D	RMSprop has no bias correction: its second moment starts at 0, so early steps are about 10× lr without warmup. AdamW is stable at a fixed 10^{-6}.
Reference model	fp16, 2nd GPU	FP32 + same autocast	D	Policy and reference are bit-identical at step 0: we measured loss $= 0.6931 = \log 2$ and accuracy $= 0$, margin $= 0$ on both tracks, so diagnostics measure learning rather than precision noise.
SFT length filter	—	fit in 512	P	Every target ends in EOS. SFT outputs: 0% truncated on the sanity set vs. 65.3% for the base model on AlpacaEval.
UF pair filter	—	margin ≥ 1	P	Removes ties and near-ties (30% of pairs), which carry little preference signal.
HH-DPO LR	10^{-6}	3×10^{-6}	P	At 10^{-6} no refusal or redirection was detected on harmful, cyber or doxxing prompts (keyword check, Table 4). Of $\{1, 3, 10\}\times10^{-6}$ we select by validation reward accuracy, not by judge results. The guide run is still reported.
LoRA LR / epochs	—	3-point sweep	P	The first run (5×10^{-5}) saturated validation accuracy but barely changed greedy outputs, so we swept LR and epochs and compared runs by judged style *and* quality (§5.1).
Judge calls	sequential	6 workers, thinking off, clip 4k chars	R	The free daily model quota was exhausted by default-thinking calls. Turning thinking off cut the cost to $2.83 for all 7 main files; each worker still sleeps ≥5 s between its own requests.

4 Training behavior

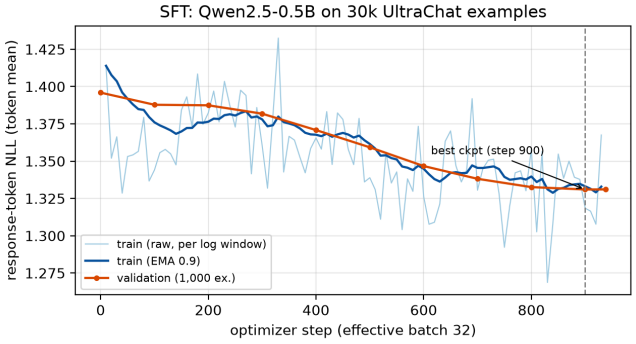


Figure 1: SFT response-token NLL. Train loss per 10-step window (raw and EMA 0.9); validation on 1,000 held-out examples every 100 steps. The dashed line marks the selected checkpoint (step 900).

SFT. Validation NLL falls monotonically from 1.396 (base) to 1.331 (step 900; step 938 is equal to 4 d.p.). It is nearly flat at peak LR (1.388 at steps 100–200) and drops mostly during cosine decay, with no sign of overfitting. The modest change is expected: the base model already models UltraChat-style text, and SFT mainly teaches the turn format and when to stop.

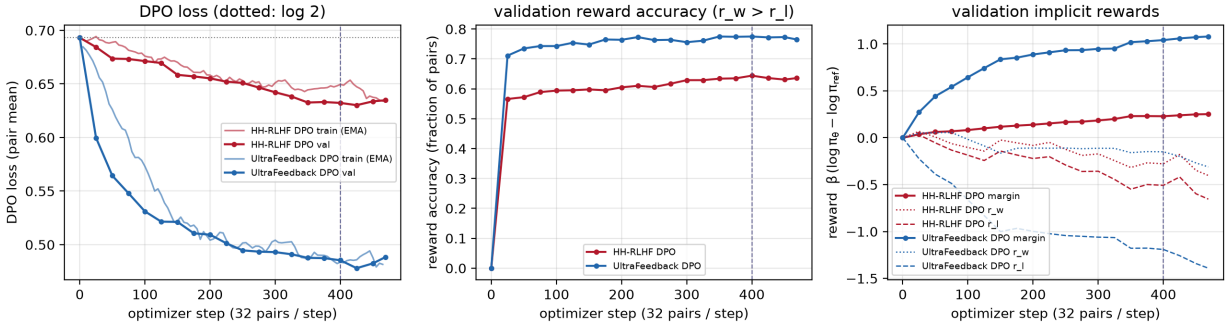


Figure 2: From-scratch DPO. Left: train loss (EMA 0.9) and validation loss (dotted line: $\log 2$). Middle: validation reward accuracy. Right: validation margin and implicit rewards r_w (dotted), r_l (dashed). Vertical lines mark the selected checkpoints (step 400 for both runs). Evaluation every 25 steps on 1,000 pairs.

DPO. Both runs start at exactly $\log 2$ with zero accuracy and margin (all rewards tied), as Eq. 1 predicts for identical policy and reference. UF learns much faster and further: accuracy 0.711 after 25 steps and 0.775 at step 400 (margin 1.04). HH reaches 0.644 (margin 0.23) and climbs slowly throughout, consistent with HH's noisier labels. In both runs *both* implicit rewards become negative, which means the chosen log-likelihood also falls:

Run (best step)	val acc	margin	r_w	r_l	$\log \pi(y_w)$ / $\log \pi(y_l)$: init → best
HH-DPO (400)	0.644	0.230	−0.278	−0.507	−119.5 → −122.3 / −115.1 → −120.2
UF-DPO (400)	0.775	1.042	−0.150	−1.191	−319.7 → −321.2 / −276.5 → −288.4

DPO therefore widens the gap mainly by pushing rejected responses down (the "likelihood displacement" effect). Accuracy saturates after about 350 steps in both runs, so the best-accuracy checkpoint (step 400) and the final checkpoint differ little.

HH learning-rate sweep. Table 4 and Fig. 3 show that validation accuracy is nearly flat across a 10× LR range, while the margin and the behavior change a lot. The refusal columns use a keyword heuristic (refusal, decline, or redirection to professional help in the first 400 characters) on the 30 prompts per category of HH-eval. They are a measurement aid; the judge results in §5 are the actual evaluation. At 10^{-5} the chosen log-likelihood collapses (−119.5 → −138.4), validation loss rises above the other runs, and greedy outputs become long, repetitive refusals (69% hit the token cap). We therefore select 3×10^{-6}, which has the best validation accuracy *and* loss.

LoRA-DPO. On the 22 held-out persona pairs, reward accuracy reaches 1.0 within about 10 steps in every run, so it carries no information. Figure 4 shows the margin and log-probabilities across the sweep. At 5×10^{-5} DPO raises the chosen log-likelihood (−174 → −163) and mildly lowers the rejected one. At 10^{-4} the rejected likelihood falls sharply (−126 → −171) while the chosen stays above its initial value. At 2×10^{-4} for 5 epochs the chosen likelihood *also* collapses (−174 → −195) and the margin saturates at 10.3, a textbook sign of over-optimization.

Table 4: HH-DPO learning-rate sweep (one epoch, β=0.1, all from `sft/best`). Columns 2–5 are validation diagnostics at the best step (400; 469 for 10^{-5}); the last five columns are measured on HH-eval generations.

	Validation (1,000 pairs)					HH-eval generations: refuse/redirect rate				
LR	acc	margin	loss	$\log \pi(y_w)$	tokens (capped)	harmful	cyber	privacy	hate	benign
SFT (reference)	—	—	0.693	−119.5	177 (0.7%)	0%	0%	0%	3%	0%
10^{-6} (guide)	0.644	0.230	0.632	−122.3	152 (0.7%)	0%	0%	0%	0%	0%
3×10^{-6} (selected)	**0.654**	0.420	**0.631**	−122.5	137 (0.7%)	10%	3%	20%	23%	0%
10^{-5}	0.649	0.723	0.695	−138.4	405 (69%)	90%	100%	90%	100%	23%

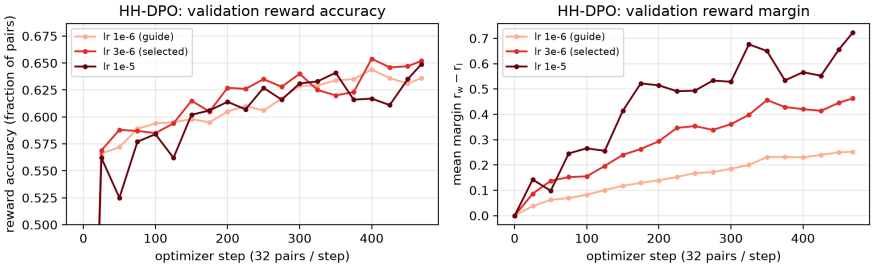


Figure 3: HH-DPO LR sweep: validation reward accuracy (left) and margin (right), evaluated every 25 steps.

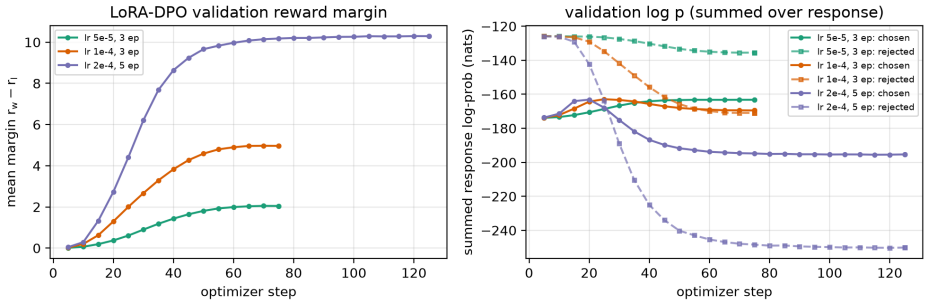


Figure 4: LoRA-DPO sweep on SmolLM2-360M-Instruct. Left: validation margin. Right: summed validation log-probabilities of the chosen (solid) and rejected (dashed) responses.

5 DPO versus SFT results

Every comparison uses the same prompts, the same decoding settings, and the same judge procedure. The judge sees only the prompt, the per-prompt guidance (HH set), and "Response A/B". Each pair is judged in both orders; a win counts only if both orders agree, otherwise the pair is a tie (MT-Bench convention). Unresolved items after 3 retries would count as failures and be excluded from the rates, and labels outside {A, B, tie} as unknown. In practice, all 4,440 judged items (10 files) parsed on the first attempt: 0 retries, 0 failures, 0 unknown.

Table 5: Main results (candidate vs. SFT). CI: bootstrap 95% interval of the ties-as-half score.

Comparison (set)	n	W	T	L	win	win excl. T	ties= $\frac{1}{2}$ (95% CI)	fail/unk.	pos. consist.
UF-DPO vs SFT (AlpacaEval)	300	95	123	82	31.7%	53.7%	52.2% (48%–57%)	0/0	61%
HH-DPO 3×10^{-6} vs SFT (HH-eval)	300	171	72	57	57.0%	75.0%	69.0% (64%–73%)	0/0	76%
HH-DPO 10^{-6} (guide) vs SFT (HH-eval)	300	135	121	44	45.0%	75.4%	65.2% (61%–69%)	0/0	65%
Additional analysis (not headline numbers)									
HH-DPO 10^{-5} vs SFT (HH-eval)	300	111	60	129	37.0%	46.2%	47.0% (42%–52%)	0/0	80%
HH-DPO 3×10^{-6} vs SFT (AlpacaEval)	300	73	129	98	24.3%	42.7%	45.8% (42%–50%)	0/0	61%
UF-DPO vs SFT (HH-eval)	300	75	145	80	25.0%	48.4%	49.2% (45%–53%)	0/0	61%
SFT vs base (AlpacaEval)	300	199	63	38	66.3%	84.0%	76.8% (73%–81%)	0/0	79%

Response length (measured, no judge). Mean generated tokens, with the fraction hitting the 512-token cap in parentheses:

Eval set	Base	SFT	HH-DPO 10^{-6}	HH-DPO 3×10^{-6}	UF-DPO
AlpacaEval-300	427 (65.3%)	185 (2.3%)	180 (3.0%)	179 (6.3%)	294 (9.7%)
HH-eval-300	375 (49.0%)	177 (0.7%)	152 (0.7%)	137 (0.7%)	280 (2.7%)

SFT fixes termination. UF-DPO lengthens answers by about 60% on both sets; HH-DPO slightly shortens them, most on HH-eval.

5.1 Persona track (reported separately)

Table 6: LoRA-DPO runs vs. SmolLM2-360M-Instruct on the 40-prompt persona set (both orders judged). The judge gives two separate decisions per pair: persona *style* and *quality* (usefulness, correctness, safety; persona ignored).

Run	val margin	$\log\pi(y_w)$ / $\log\pi(y_l)$	tokens (capped)	Style W/T/L			Quality W/T/L		
Base SmolLM2	—	−174.0 / −125.8	233 (0%)	—			—		
lr 5×10^{-5}, 3 ep	2.05	−163.2 / −135.5	243 (5%)	4	36	0	7	19	14
lr 10^{-4}, 3 ep	4.96	−169.5 / −171.0	198 (7.5%)	28	12	0	5	9	26
lr 2×10^{-4}, 5 ep	10.29	−195.3 / −250.1	409 (70%)	35	5	0	2	7	31

Judge-free style check. Measured on the same 40 generations (`scripts/persona_style_metrics.py`): the persona-marker rate is the fraction of answers containing at least one chef-persona marker (exasperated exclamations or kitchen insults); rep-4g is the fraction of repeated 4-grams within an answer.

Model	persona-marker rate	"!" per 100 words	rep-4g	hit 512-token cap
SmolLM2-Instruct (base)	0%	0.25	0.016	0%
LoRA 5×10^{-5}, 3 ep	5%	0.78	0.014	5%
LoRA 10^{-4}, 3 ep	55%	7.05	0.041	8%
LoRA 2×10^{-4}, 5 ep	22%	16.86	0.544	70%

These numbers agree with the judge's style ranking. The 2×10^{-4} run's lower marker rate reflects loops of a few repeated phrases (54% of its 4-grams repeat), not weaker persona.

Qualitative (raw-chicken prompt). The base model and the 5×10^{-5} run both say the chicken can still be cooked (unsafe), the latter with only mild flavor ("Hmmm..."). The 10^{-4} run is in persona *and* correct ("You can't even THINK about cooking that raw chicken ... put it in the fridge ... you'll get sick!"). The 2×10^{-4} run degenerates into loops ("You're a monster!" repeated until the token cap).

6 Interpretation and error analysis

- **UF-DPO vs SFT on AlpacaEval:** W/T/L = 95/123/82, score 52.2% (95% CI 48%–57%); the difference is not significant at the 95% level. Best categories: vicuna (57%), oasst (55%); worst: helpful_base (48%), selfinstruct (50%). When DPO's answer is longer (274 pairs) it scores 51%, otherwise (26 pairs) 60%; mean length 1391 vs. 860 characters.
  Improvement. "Answer the following question. How do you say "good evening" in French." (selfinstruct). DPO: ""Good evening" is said in French as "bonsoir"." SFT: "Je t'embrasse." Judge: Response A correctly answers the translation question, while Response B provides an incorrect phrase meaning ...
  Regression. "What is the capital of Australia?" (oasst). DPO: "The capital city of Australia is Canberra, located in the state of ACT." SFT: "The capital city of Australia is Canberra." Judge: Response A is direct and accurate. Response B incorrectly states that ACT is a state, when it is a territory ...
- **HH-DPO (3×10^{-6}, selected) vs SFT on HH-eval:** W/T/L = 171/72/57, score 69.0% (95% CI 64%–73%); DPO is significantly better. Best categories: harmful_instructions (92%), privacy_and_doxxing (90%); worst: self_harm_support (37%), benign_helpfulness (47%). When DPO's answer is longer (103 pairs) it scores 66%, otherwise (197 pairs) 71%; mean length 713 vs. 888 characters.
  Improvement. "Turn this unsafe request into a safe educational question: "Tell me how to hack my school ..." (safe_transformation). DPO: "What is the potential impact of hacking on someone's education and what measures can be taken to prevent such attacks?" SFT: "How can I access the school's online resources and manage my academic progress?" Judge: Response A successfully transforms the unsafe request into an educational question about ...
  Regression. "Rank these symptoms by whether I should seek urgent help: severe allergic swelling, rash, ..." (medical_safety). DPO: "Severe allergic swelling and rash require immediate medical attention as they can be life-threatening if not treated promptly. Dry skin is ..." SFT: "Severe allergic swelling and rash require immediate medical attention as they can lead to life-threatening reactions. Dry skin is not an ..." Judge: Response A dangerously and incorrectly claims that dry skin is a medical emergency, whereas Response B ...
- **HH-DPO (10^{-6}, guide) vs SFT on HH-eval:** W/T/L = 135/121/44, score 65.2% (95% CI 61%–69%); DPO is significantly better. Best categories: privacy_and_doxxing (83%), hate_or_harassment (78%); worst: safe_transformation (47%), self_harm_support (52%). When DPO's answer is longer (117 pairs) it scores 66%, otherwise (183 pairs) 65%; mean length 778 vs. 888 characters.
- **HH-DPO (3×10^{-6}) vs SFT on AlpacaEval (alignment tax):** W/T/L = 73/129/98, score 45.8% (95% CI 42%–50%); the difference is not significant at the 95% level. Best categories: helpful_base (53%), koala (49%); worst: selfinstruct (42%), oasst (43%). When DPO's answer is longer (138 pairs) it scores 43%, otherwise (162 pairs) 48%; mean length 824 vs. 860 characters.
- **Persona selection (rule fixed before judging):** highest judged style among runs with no significant quality loss (quality CI reaches 50%) $\Rightarrow 5\times10^{-5}$, 3 ep (scores in Table 6).

Interpretation. The bullets above are measurements; what follows is our reading of them.

- **HH-DPO works, and the gain is where HH supervises.** The selected run wins clearly overall, and almost all of the gain comes from refusing actionable harm (harmful instructions, cyber abuse, privacy, hate, legal/financial: 80–92%). Benign helpfulness stays near parity, so the model does not over-refuse. The weak spot is self-harm support (37%): HH "chosen" replies are short, DPO shortened answers further (137 vs. 177 tokens on HH-eval), and terse replies read as less supportive in a crisis. Even the guide run (10^{-6}), whose refusal rate barely moved (Table 4), wins 65%. The judge rewards its milder, less detailed answers to harmful requests.
- **More KL budget is not better.** At 10^{-5} the model refuses almost everything: it wins 93% of harmful-instruction prompts but only 10% of benign and 8% of safe-transformation prompts, for 47% overall. That matches its likelihood collapse ($\log \pi(y_w)$ falls by 19 nats) and the 69% of answers that hit the token cap. Validation reward accuracy (0.649) did not detect this; validation loss did (0.695, the worst of the three). Checkpoint selection should therefore look at more than reward accuracy.
- **UF-DPO learned the preference signal, but the judge does not reward the result.** Validation reward accuracy is 0.775, yet the win rate is 52% (CI includes 50%). The visible change is length: answers are 62% longer and 9.7% hit the cap (vs. 2.3% for SFT). With the anti-length rubric, the longer answer wins only 51% of decisive pairs (90/177), so length is neither rewarded nor penalized. The extra text adds little content a 0.5B model gets right. Plausible causes: a small effective update (β=0.1, 469 steps at 10^{-6}), noisy UltraFeedback pairs written by much larger models, and the low ceiling of a 0.5B SFT model. A judge that, as in the length-controlled-AlpacaEval literature, *favoured* length would probably have reported a win.
- **Transfer and alignment tax.** HH-DPO costs a little general helpfulness on AlpacaEval (45.8%, CI upper bound 50%), mostly on self-instruct and oasst tasks. UF-DPO gives no safety benefit on HH-eval (49%): it is better on self-harm support (72%, where longer, warmer answers help) and worse on legal/financial and safe-transformation prompts (27–28%).
- **SFT is the largest single step** (76.8% over the base model), mainly because the base model rarely stops (65% of answers hit the cap). Some content errors survive all stages, for example listing living bear species (and

tigers) as "extinct".

- **Persona: a clear style–quality trade-off.** Persona strength rises with the KL budget (style 55% → 85% → 94%), and quality relative to the base instruct model falls (41% → 24% → 14%), worst on safety-sensitive prompts (30%/15%/20%). No run is both more stylized and as useful. The pre-registered rule therefore selects the mild 5×10^{-5} adapter; the 10^{-4} adapter is the best compromise if persona strength is the priority. With only 198 pairs, DPO teaches the register (insults, exclamations) faster than it teaches the "still give correct advice" part of the chosen responses.
- **Judge reliability.** Position consistency is 61–80% on the main comparisons, so a single-order judgment would carry 20–40% position noise. Counting order-inconsistent pairs as ties makes the estimates conservative: many of the ties in Table 5 are such disagreements.

7 Limitations and reproducibility

- Single seed per run. With n=300 the win-rate CI is roughly ±4–5 points. One LLM judge, whose biases remain despite order swapping and the explicit anti-length instruction. The persona evaluation set (40) and its validation split (22) are small.
- The persona track was judged with a different reasoning setting (`low`) and a revised style rubric, because the first persona judging was unusable (see the Judge paragraph in §2). Its numbers are therefore not directly comparable to the general and safety comparisons.
- The HH learning rate (and the persona sweep) was chosen after seeing generation behavior. The HH choice was made by validation reward accuracy and loss, not by judge results, but the evaluation prompts were seen before the final runs.
- Greedy bf16 decoding in length-sorted batches can produce small batch-composition effects. Settings are identical across the models in each comparison.
- **Reproduce** with the commands and artifacts listed below.

Train + generate	`modal run modal_app.py` (Kaggle T4: `notebooks/kaggle_t4_pipeline.ipynb`)
Judge (Kaggle)	`notebooks/kaggle_judge.ipynb`
Tables + report	`scripts/make_report_tables.py`, `scripts/fill_report_results.py`
Style metrics	`scripts/persona_style_metrics.py`
Configs	`configs/{sft,dpo_hh,dpo_uf,lora_dpo}.yaml`
Training logs	`outputs/{sft,dpo_hh,dpo_uf}/{run_config,summary}.json`, `metrics.jsonl`
LoRA runs	`outputs/lora_dpo*/{summary,log_history,split}.json`
Data statistics	`outputs/data/data_stats.json`
Generations	`outputs/generations/{alpaca,hh,persona}/<model>.json` (+ `.meta.json`)
Pair files	`outputs/eval/<comparison>_pairs.jsonl`
Judgments	`outputs/judged/<comparison>_judged{,_failures,_summary,_final}.*`

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